

Solving analytical queries on Redshift Cluster

1. Top 10 ATMs where most transactions are in the 'inactive' state

```
SELECT
  a.atm_number,
  a.atm_manufacturer,
  l.location,
  count(f.trans_id) as total_transaction_count,
  SUM(CASE WHEN f.atm_status = 'Inactive'
  THEN 1
  ELSE 0
  END) as inactive_count,
  (inactive_count/total_transaction_count)*100 as inactive_count_percent
FROM
  etl_atm_data.DIM_ATM as a,
  etl_atm_data.FACT_ATM_TRANS as f,
  etl_atm_data.DIM_LOCATION as l
WHERE
  a.atm_id = f.atm_id AND a.atm_location_id = l.location_id
GROUP BY
  a.atm_number,
  a.atm_manufacturer,
  l.location
ORDER BY
  inactive_count DESC
LIMIT 10;
```

atm_number ▾	atm_manufacturer ▾	location ▾	total_transaction_count ▾	inactive_count ▾	inactive_count_percent ▾
16	NCR	Skive Lobby	44043	44043	100
12	NCR	Årøester ÅV Duus	33982	33982	100
2	NCR	Vejgaard	33725	33725	100
88	NCR	Storcenter indg. A	32183	32183	100
30	NCR	Nyk Å, bing Mors	30883	30883	100
52	NCR	Fars Å, _	27361	27361	100
50	NCR	Aarhus	23416	23416	100
29	NCR	Skelagervej 15	20773	20773	100
81	NCR	Spar K Å, bmand Tornh Å, j	20148	20148	100
102	NCR	Aalborg Storcenter Afd	18297	18297	100

2. Number of ATM failures corresponding to the different weather conditions recorded at the time of the transactions

```
WITH failed_atm AS
(SELECT
  weather_main,
  count(trans_id) as total_transaction_count,
  SUM(CASE WHEN atm_status = 'Inactive'
    THEN 1
    ELSE 0
  END) as inactive_count
FROM
  etl_atm_data.FACT_ATM_TRANS
WHERE
  weather_main != ''
GROUP BY
  weather_main)
SELECT *,
  ROUND(CAST(inactive_count AS numeric(10,2))/ total_transaction_count*100,2) AS
  inactive_count_percent
FROM
  failed_atm
ORDER BY
  inactive_count_percent DESC;
```

weather_main	total_transaction_count	inactive_count	inactive_count_percent
Snow	23405	4813	20.56
Fog	18174	3729	20.52
Clouds	1181901	194027	16.42
Rain	545135	86017	15.78
Clear	543949	85531	15.72
Mist	82801	12864	15.54
Thunderstorm	2549	361	14.16
Drizzle	62530	8670	13.87
TORNADO	38	1	2.63
Haze	3	0	0.00

3. Top 10 ATMs with the most number of transactions throughout the year

```
SELECT
    a.atm_number,
    a.atm_manufacturer,
    l.location,
    count(f.trans_id) as total_transaction_count
FROM
    etl_atm_data.DIM_ATM as a,
    etl_atm_data.FACT_ATM_TRANS as f,
    etl_atm_data.DIM_LOCATION as l
WHERE
    a.atm_id = f.atm_id AND a.atm_location_id = l.location_id
GROUP BY
    a.atm_number,
    a.atm_manufacturer,
    l.location
ORDER BY
    total_transaction_count DESC
LIMIT 10;
```

atm_number	atm_manufacturer	location	total_transaction_count
39	NCR	Svenstrup	55380
20	NCR	Bispensgade	54211
10	NCR	NÅfÅ_rresundby	53794
24	NCR	Hobro	53378
45	NCR	Abildgaard	53198
16	NCR	Skive Lobby	44043
40	Diebold Nixdorf	Frederikshavn	43767
1	NCR	NÅfÅ'stved	42787
41	Diebold Nixdorf	Skagen	42732
48	Diebold Nixdorf	Intern BrÅfÅ_nderslev	42493

4. Number of overall ATM transactions going inactive per month for each month

```
WITH failed_atm_monthly AS (
  SELECT
    d.year,
    d.month,
    count(f.trans_id) as total_transaction_count,
    SUM(CASE WHEN f.atm_status = 'Inactive'
      THEN 1
      ELSE 0
    END) as inactive_count
  FROM
    etl_atm_data.FACT_ATM_TRANS as f,
    etl_atm_data.DIM_DATE as d
  WHERE
    f.date_id = d.date_id
  GROUP BY
    d.year,
    d.month)
SELECT *,
  ROUND(CAST(inactive_count AS numeric(10,2))/ total_transaction_count*100,2) AS
  inactive_count_percent
FROM
  failed_atm_monthly
ORDER BY
  month;
```

year	month	total_transaction_count	inactive_count	inactive_count_percent
2017	April	218865	41830	19.11
2017	August	217218	36713	16.90
2017	December	197048	20476	10.39
2017	February	182659	36656	20.07
2017	January	180195	35953	19.95
2017	July	227682	38139	16.75
2017	June	225166	36789	16.34
2017	March	209586	41046	19.58
2017	May	222418	37679	16.94
2017	November	193967	21684	11.18
2017	October	191667	21780	11.36
2017	September	202101	28913	14.31

5. Top 10 ATMs with the highest total withdrawn amount throughout the year

```
SELECT
  a.atm_number,
  a.atm_manufacturer,
  l.location,
  SUM(f.transaction_amount) as total_amount_withdrawn
FROM
  etl_atm_data.DIM_ATM as a,
  etl_atm_data.FACT_ATM_TRANS as f,
  etl_atm_data.DIM_LOCATION as l
WHERE
  a.atm_id = f.atm_id
  AND a.atm_location_id = l.location_id
GROUP BY
  a.atm_number,
  a.atm_manufacturer,
  l.location
ORDER BY
  total_amount_withdrawn DESC
LIMIT 10;
```

Result 1 (10)					Export
☐	atm_number	atm_manufacturer	location	total_amount_withdrawn	
☐	39	NCR	Svenstrup	277097637	
☐	20	NCR	Bispensgade	271008803	
☐	24	NCR	Hobro	268289882	
☐	10	NCR	NÅfÅ, resundby	267379103	
☐	45	NCR	Abildgaard	265639616	
☐	16	NCR	Skive	220677013	
☐	40	Diebold Nixdorf	Frederikshavn	219812287	
☐	41	Diebold Nixdorf	Skagen	214127315	
☐	1	NCR	NÅfÅ, stved	213721117	
☐	48	Diebold Nixdorf	BrÅfÅ, nderslev	212883099	

6. Number of failed ATM transactions across various card types

```
SELECT
  ct.card_type,
  COUNT(trans_id) as total_transaction_count,
  SUM(case when atm_status = 'Inactive' then 1 else 0 end) as inactive_count
FROM
  etl_atm_data.fact_atm_trans f,
  etl_atm_data.dim_card_type ct
WHERE
  f.card_type_id = ct.card_type_id
GROUP BY
  ct.card_type;
```

Result 1 (12)			
☐	card_type	total_transaction_count	inactive_count
☐	Dankort	28581	4557
☐	Visa Dankort - on-us	748805	112972
☐	Visa Dankort	427840	60547
☐	VISA	170828	30713
☐	HÃfÃ\vekort - on-us	62487	10331
☐	HÃfÃ\vekort	8459	1208
☐	CIRRUS	17362	2953
☐	VisaPlus	1134	150
☐	Mastercard - on-us	458226	86000
☐	MasterCard	400507	63482
☐	Dankort - on-us	143813	24680
☐	Maestro	530	65

7. Number of transactions happening on an ATM on weekdays and on weekends throughout the year. Order this by the ATM_number, ATM_manufacturer, location, weekend_flag and then total_transaction_count

```
SELECT
  a.atm_number,
  a.atm_manufacturer,
  l.location,
  CASE
    WHEN d.weekday IN ('Saturday', 'Sunday') THEN 1
    ELSE 0
  END AS weekend_flag,
  COUNT(f.trans_id) AS total_transaction_count
FROM
  etl_atm_data.fact_atm_trans f
JOIN
  etl_atm_data.dim_atm a ON f.atm_id = a.atm_id
JOIN
  etl_atm_data.dim_location l ON a.atm_location_id = l.location_id
JOIN
  etl_atm_data.dim_date d ON f.date_id = d.date_id
GROUP BY
  a.atm_number,
  a.atm_manufacturer,
  l.location,
  weekend_flag
ORDER BY
  a.atm_number,
  a.atm_manufacturer,
  l.location,
  weekend_flag,
  total_transaction_count DESC
LIMIT 10;
```

Result 1 (10)					Export ▾	Chart
☐	atm_number	atm_manufacturer	location	weekend_flag	total_transaction_count	
☐	1	NCR	NÃfÃstved	0	32711	
☐	1	NCR	NÃfÃstved	1	10076	
☐	10	NCR	NÃfÃrresundby	0	41667	
☐	10	NCR	NÃfÃrresundby	1	12127	
☐	100	NCR	Intern Skive	0	17812	
☐	100	NCR	Intern Skive	1	1	
☐	101	NCR	Bryggen Vejle	0	11693	
☐	101	NCR	Bryggen Vejle	1	3247	
☐	102	NCR	Aalborg Storcenter Afd	0	14556	
☐	102	NCR	Aalborg Storcenter Afd	1	3741	

8. Most active day in each ATMs from location "Vejgaard"

```
WITH daily_transactions AS (  
  SELECT  
    a.atm_number,  
    a.atm_manufacturer,  
    l.location,  
    d.weekday,  
    COUNT(f.trans_id) AS transaction_count  
  FROM  
    etl_atm_data.fact_atm_trans f  
  INNER JOIN  
    etl_atm_data.dim_atm a ON f.atm_id = a.atm_id  
  INNER JOIN  
    etl_atm_data.dim_location l ON a.atm_location_id = l.location_id  
  INNER JOIN  
    etl_atm_data.dim_date d ON f.date_id = d.date_id  
  WHERE  
    l.location = 'Vejgaard'  
  GROUP BY  
    a.atm_number,  
    a.atm_manufacturer,  
    l.location,  
    d.weekday),  
ranked_transactions AS (  
  SELECT *,  
    ROW_NUMBER() OVER (PARTITION BY atm_number ORDER BY transaction_count  
DESC) AS rank  
  FROM  
    daily_transactions)  
SELECT  
  atm_number,  
  atm_manufacturer,  
  location,  
  weekday,  
  transaction_count  
FROM  
  ranked_transactions  
WHERE  
  rank = 1;
```


Result 1 (2)					Export ▾	● Chart
☐	atm_number	atm_manufacturer	location	weekday	transaction_count	
☐	2	NCR	Vejgaard	Friday	6290	
☐	103	Diebold Nixdorf	Vejgaard	Friday	4757	